

CREDIT EDA CASE STUDY

Business objectives:

- This case study aims to identify patterns which indicate if a client has difficulty paying their instalments which may be used for taking actions such as denying the loan , reducing the amount loan, lending at a higher interest rate etc. this will ensure that the consumers capable of repaying the loan are not rejected . identification of such applicants using EDA is the aim of this case study.
- The company wants to understand the driving factors behind loan default,i.e. the variables which are strong indicators of default. The company can utilise this knowledge for its portfolio and risk assessment.

PROBLEM STATEMENT

The data which have analysed contains the information about loan application at the time of applying the loan.it contains two types of scenarios:

- The client with payment difficulties: he/she had late payment more than X days on at least one of the first Y instalment of the loan in our sample.(in our analysis,it is mentioned as target=1)
- All other cases: all other cases when payment is paid on time (in our analysis, it is mentioned as target=0)

ANALYSIS AND STEPS TAKEN

1] data sourcing(already provided in assignment)

2. Data loading and data cleaning:

- Fixing the rows and column
- Handling and removing missing columns

3] univariate analysis:

- Categorical- categorical analysis
- Numerical –numerical analysis

4] bivariate analysis

UNIVARIATE ANALYSIS

distribution of income type:

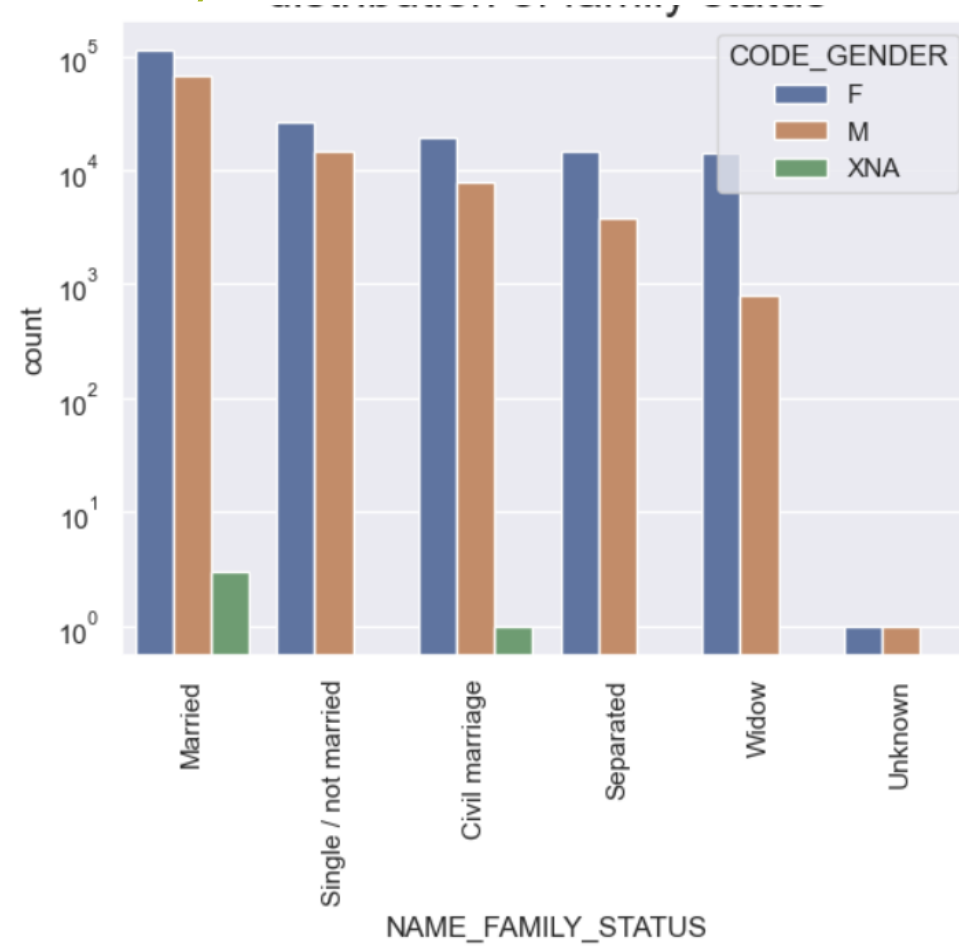


#as you can see females are having more credits than male

#for working, commercial association, pensioner , state servant the number of credits are higher than other

#for students , unemployed, businessman, maternity leave the number of credits are less

Distribution of family status:



married are having higher credit

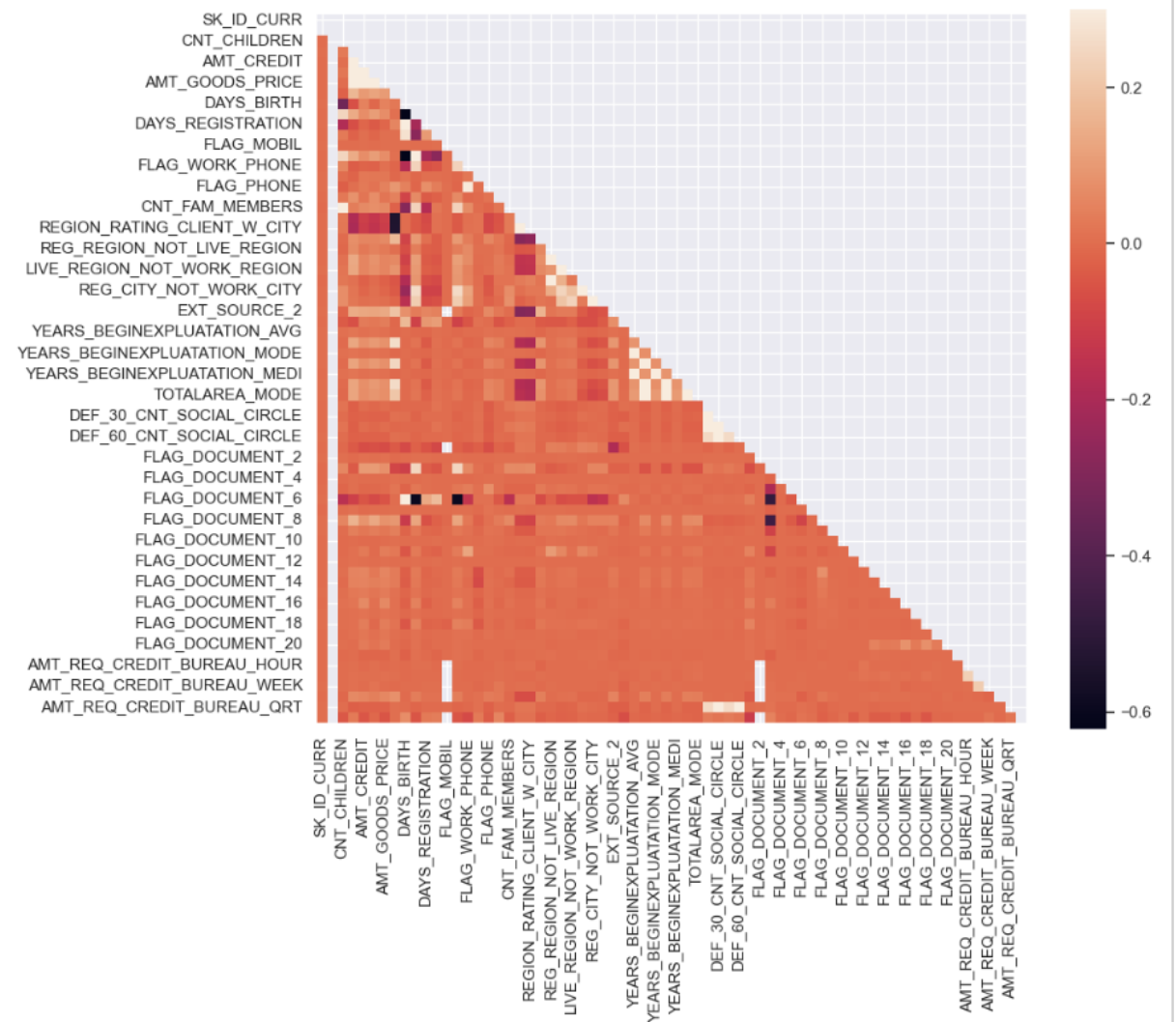
Distribution of income range :



- 1]female counts are higher than male .
- 2]income range from 100000 to 200000 is having more number of credits.
- 3]this graph shows that females are more than male in having credits for that range.
- 4]very less count for income range 400000 and above

Plotting the correlation with heatmap with target variable o that is all other cases when payment is paid on time

- 1]credit amount is inversely proportional to the date of birth, which means credit amount is higher for low age and vice-versa.
- 2]credit amount is inversely proportional to the number of children client have, means credit amount is higher for less children count have and vice-versa
- 3]less children client have in densely populated area.
- 4]the income is also higher in densely populated area



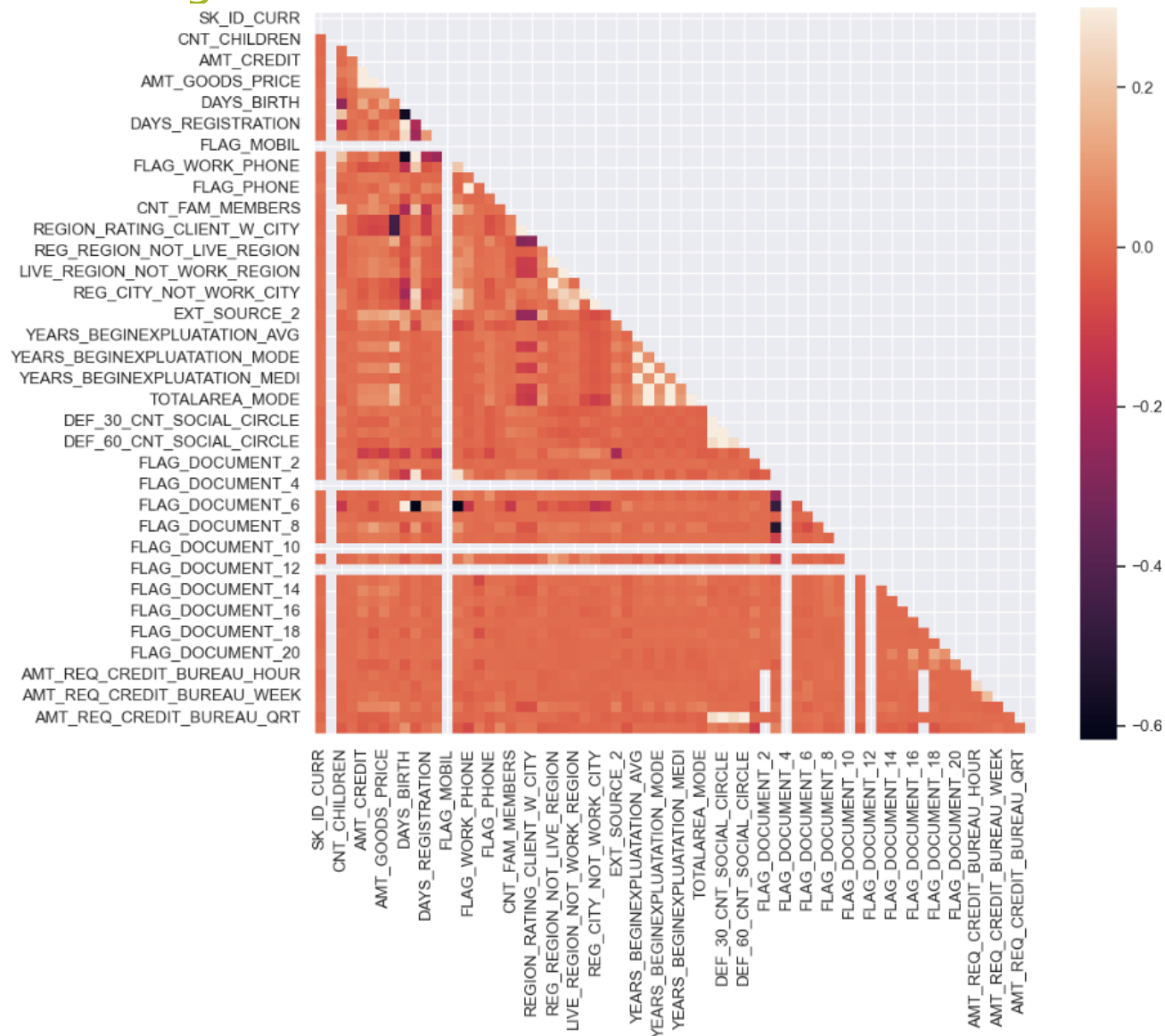
Top 10 correlation for target o

EXT_SOURCE_1	DAYS_BIRTH	0.601210
DAYS_EMPLOYED	DAYS_BIRTH	0.618048
AMT_CREDIT	AMT_ANNUITY	0.771309
AMT_GOODS_PRICE	AMT_ANNUITY	0.776686
LIVE_CITY_NOT_WORK_CITY	REG_CITY_NOT_WORK_CITY	0.830381
LIVE_REGION_NOT_WORK_REGION	REG_REGION_NOT_WORK_REGION	0.861861
CNT_FAM_MEMBERS	CNT_CHILDREN	0.878571
REGION_RATING_CLIENT_W_CITY	REGION_RATING_CLIENT	0.950149
AMT_CREDIT	AMT_GOODS_PRICE	0.987250
DAYS_EMPLOYED	FLAG_EMP_PHONE	0.999758
...		

Plotting the correlation with heatmap with target variable 1

this heat map for target1 is also having quite a same observation just like target0. but for few points are different . They are as below:

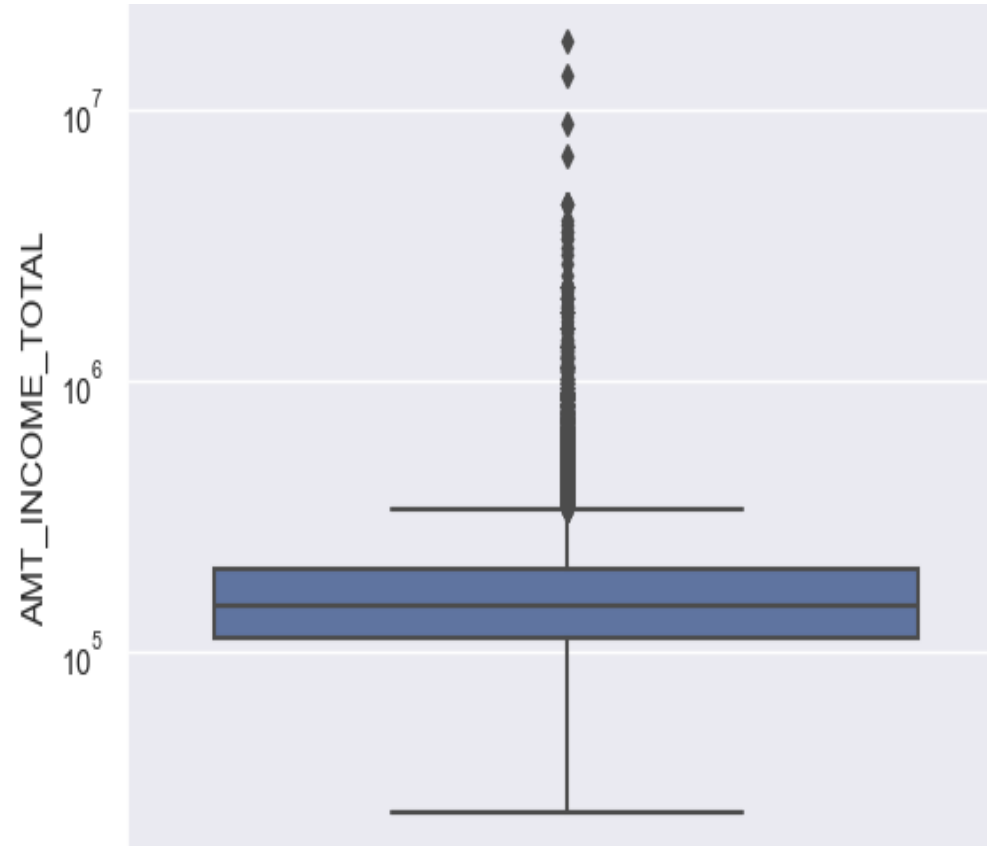
- 1] the client's permanent address does not match contact address are having less children and vice-versa
- 2] the client's permanent address does not match work address are having less children and vice-versa



Top 10 correlation for target 1

EXT_SOURCE_1	DAYS_BIRTH	0.570054
DAYS_EMPLOYED	DAYS_BIRTH	0.575097
FLAG_EMP_PHONE	DAYS_BIRTH	0.578519
AMT_CREDIT	AMT_ANNUITY	0.752195
AMT_GOODS_PRICE	AMT_ANNUITY	0.752699
REG_CITY_NOT_WORK_CITY	LIVE_CITY_NOT_WORK_CITY	0.778540
REG_REGION_NOT_WORK_REGION	LIVE_REGION_NOT_WORK_REGION	0.847885
CNT_CHILDREN	CNT_FAM_MEMBERS	0.885484
REGION_RATING_CLIENT_W_CITY	REGION_RATING_CLIENT	0.956637
AMT_CREDIT	AMT_GOODS_PRICE	0.983103
DAYS_EMPLOYED	FLAG_EMP_PHONE	0.999702

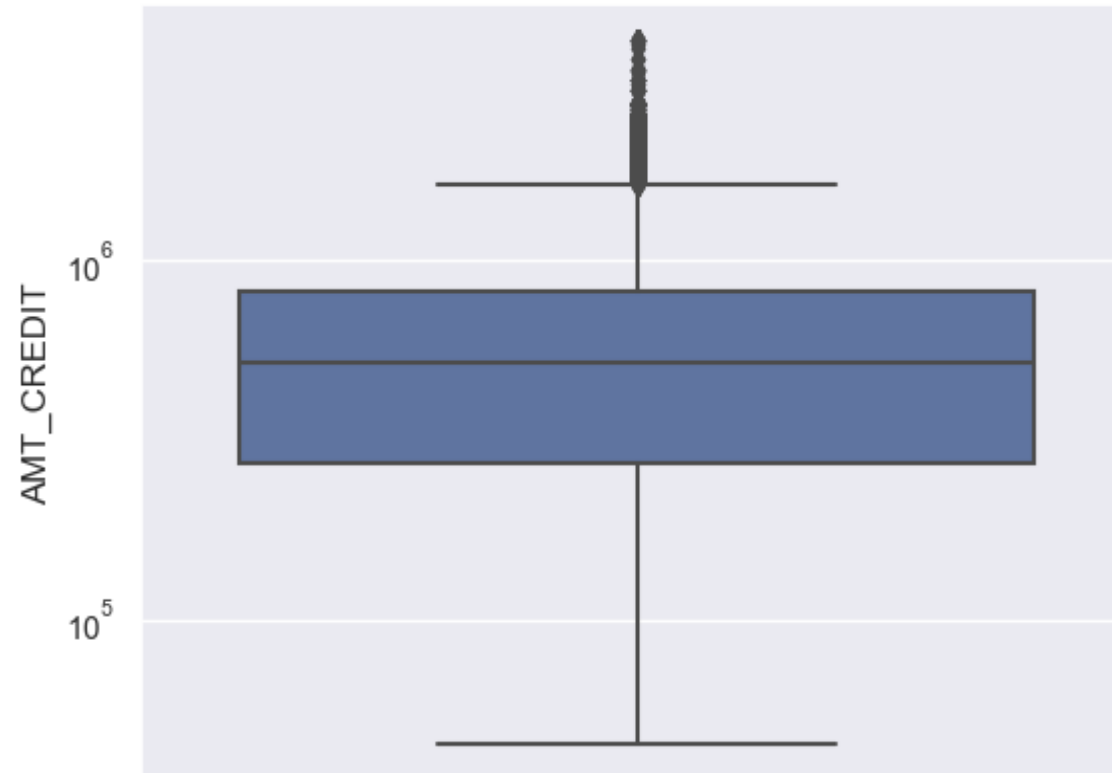
Using box plot for analysis(for target o) distribution of income amount



Few points can be concluded from the graph above:

1. Some outliers are noticed in income amount.
2. The third quartiles is very slim for income amount

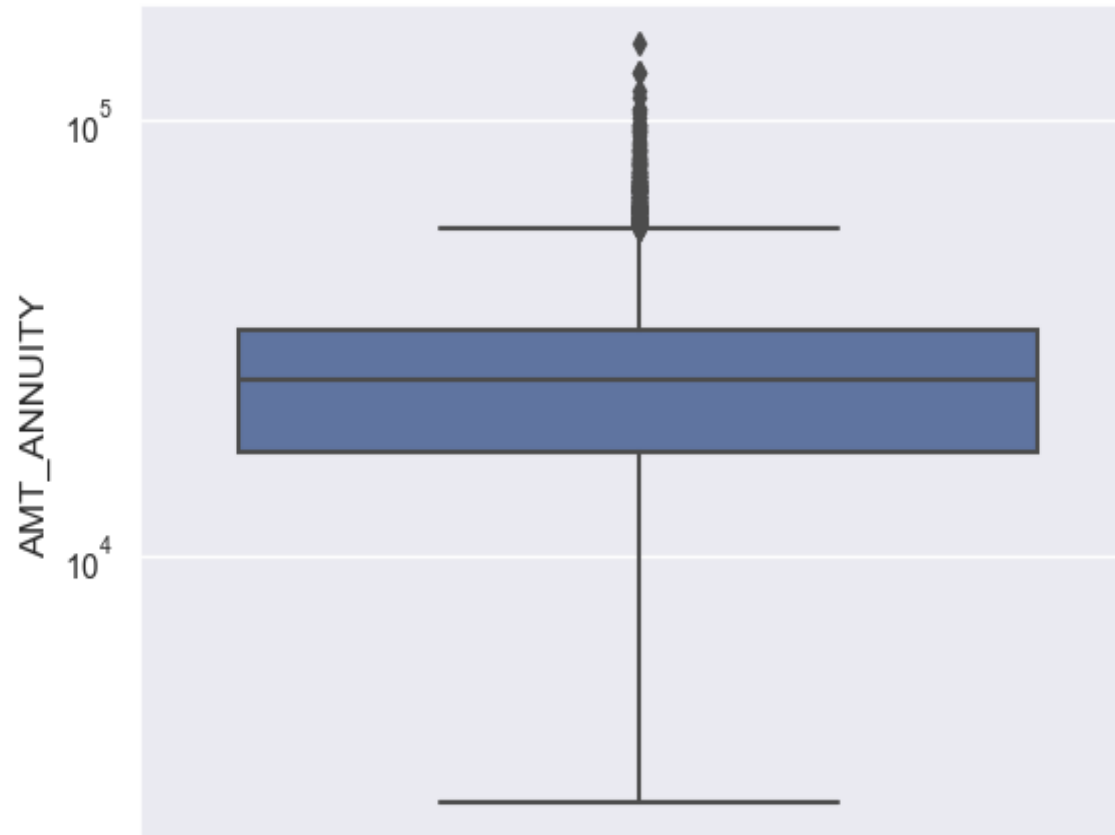
Distribution of credit amount



1. Some outliers are noticed in credit amount
2. the first quartile is bigger than third quartile for credit amount which means most of the credits of the clients are present in the first quartile.

Using box plot for analysis for target1

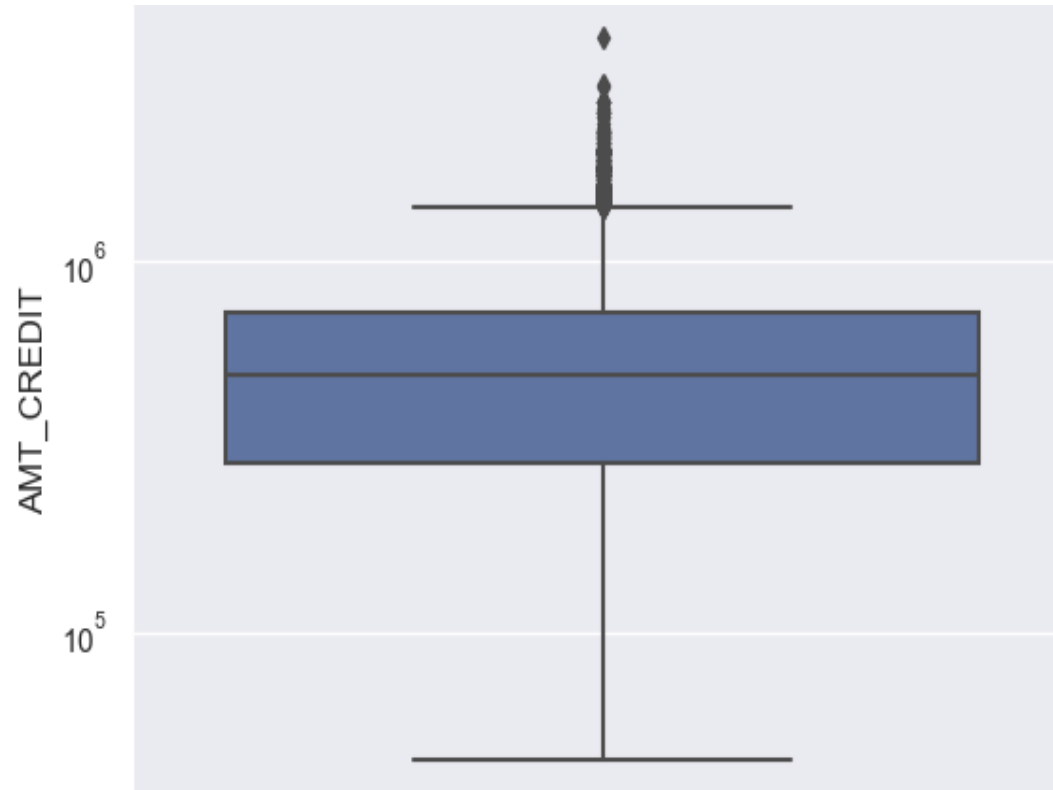
Distribution of annuity amount



1] some outliers are noticed in annuity amount.

2] the first quartile is bigger than third quartile for annuity amount which means most of the annuity clients are from first quartile.

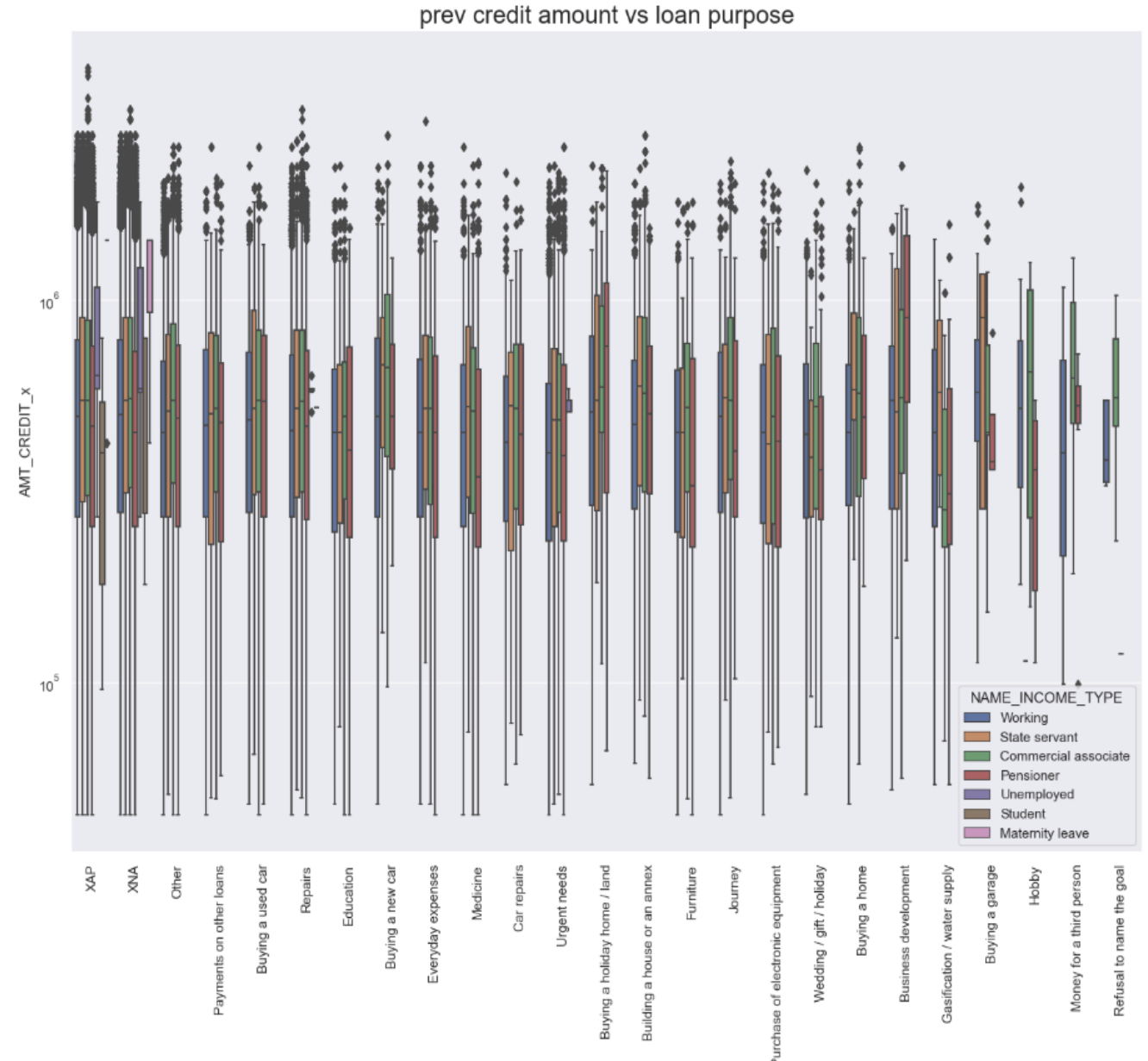
Distribution of credit amount



1. Some outliers are noticed in credit amount
2. The first quartile is bigger than third quartile for credit amount which means most of the credits are present in the first quartile

Merged the two dataset
bivariate analysis
box plotting for credit amount

- 1]the credit amount of loan purposes like buying a home, buying a land, buying a new car and buying a house is higher.
- 2]income type of state servants have a significant amount of credit applied
- 3]money for third person or a hobby is having less credits applied for



CONCLUSION

- Banks should focus more on contract type 'student', 'pensioner' and businessman with housing type other than 'co-op apartment' for successful payments.
- Banks should focus less on income type 'working' as they are having most number of unsuccessful payments
- Also with loan purpose 'repair' is having higher number of unsuccessful payments on time.
- Get as much clients from housing type 'with aparents' as they are having least number pf unsuccessful payments.